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Non-invasive Evaluation of Glucose Concentration by Measuring Ultrasound Velocity

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Research Motivation

1. Due to the recent increase in diabetic patients, research on non-invasive blood glucose measurement technology is being actively conducted.
 2. Blood glucose management standards are based on both clinical symptoms experienced and objective indicators.
 3. In studies related to ultrasound and glucose concentration, measurements are taken over a very wide range compared to human blood glucose levels.
- Fasting glucose levels are defined as 70-100mg/dL for normal, 100-125mg/dL for high-risk, and 126mg/dL or higher for diabetes.



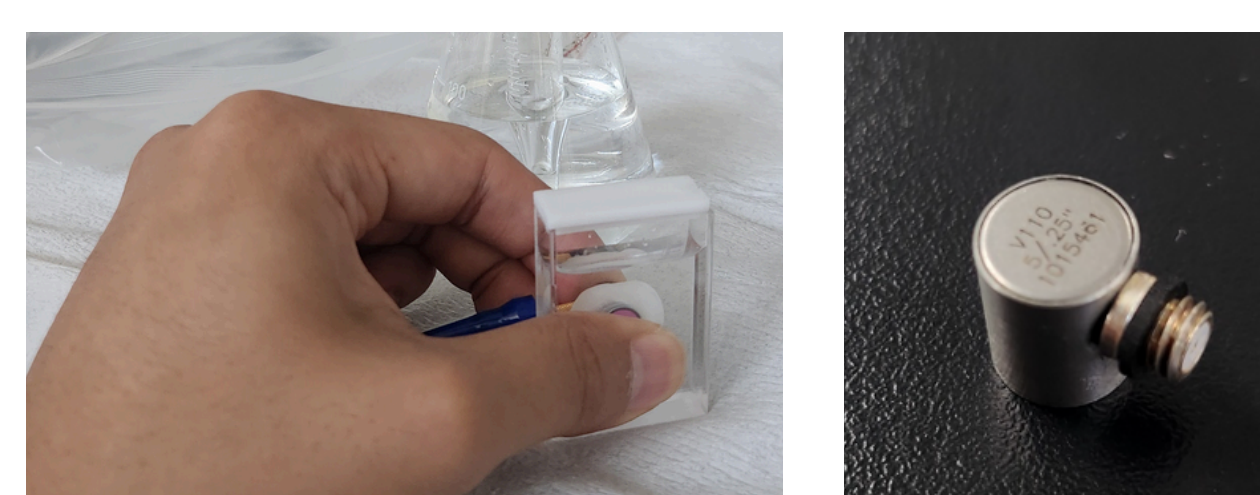
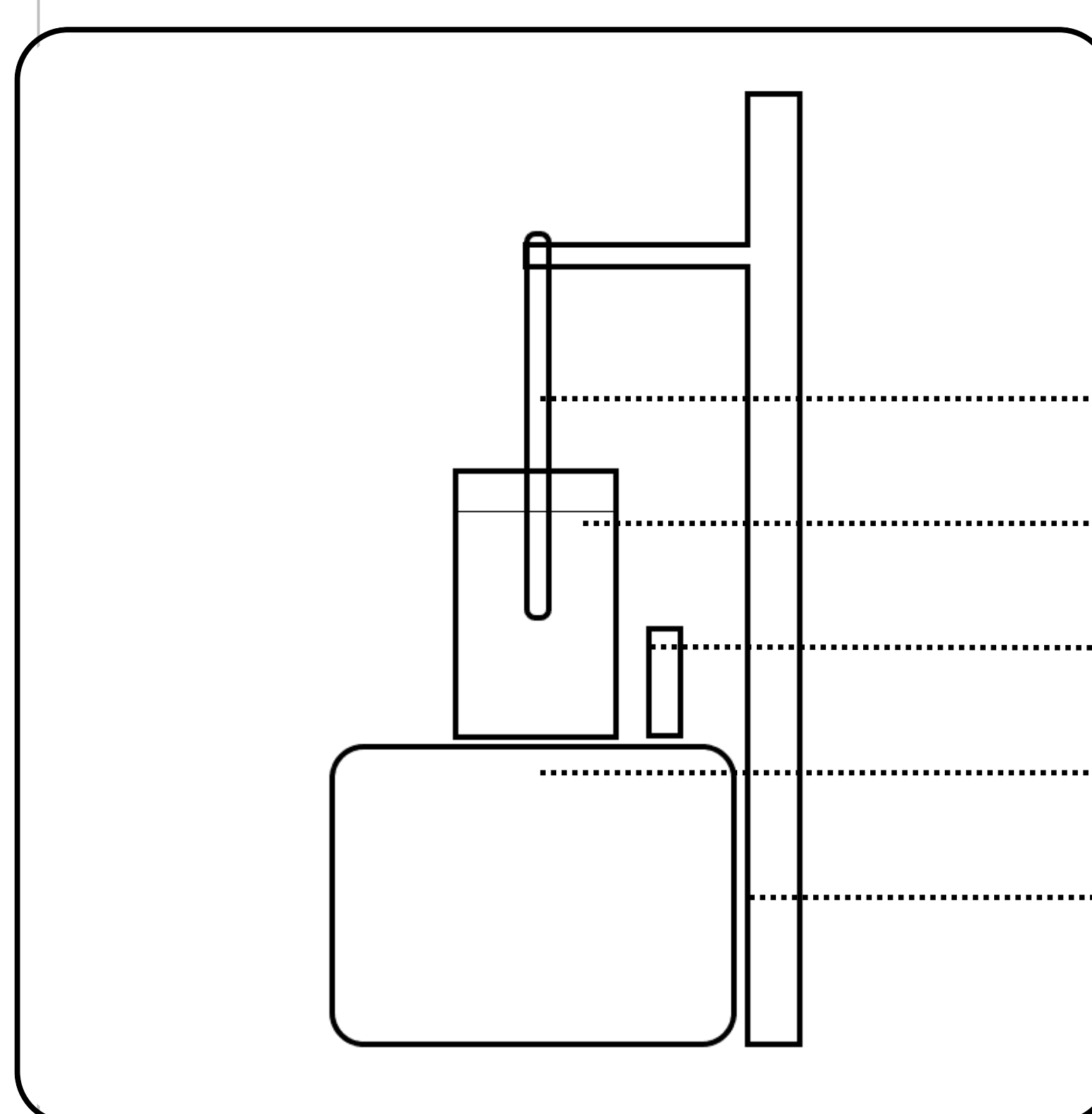
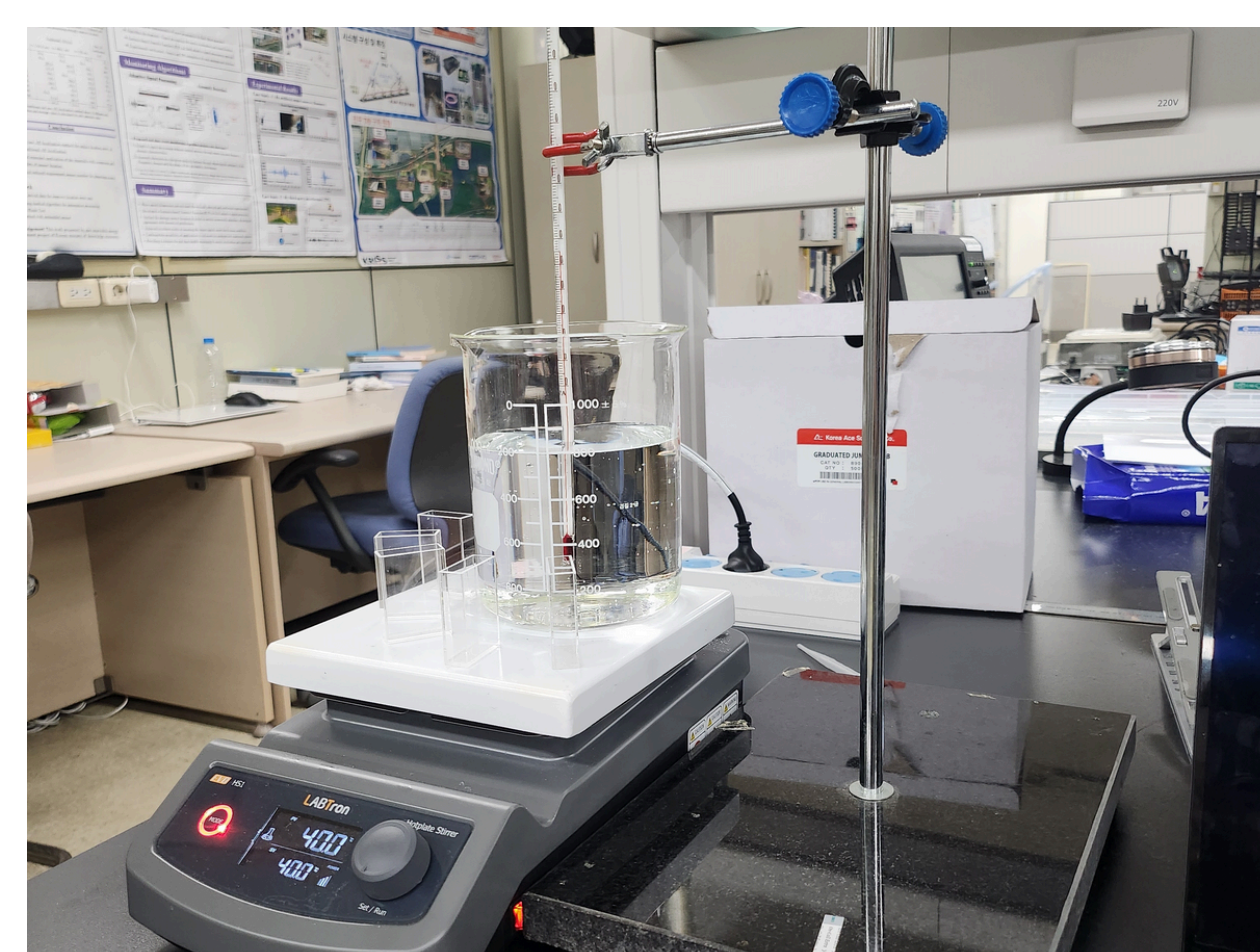
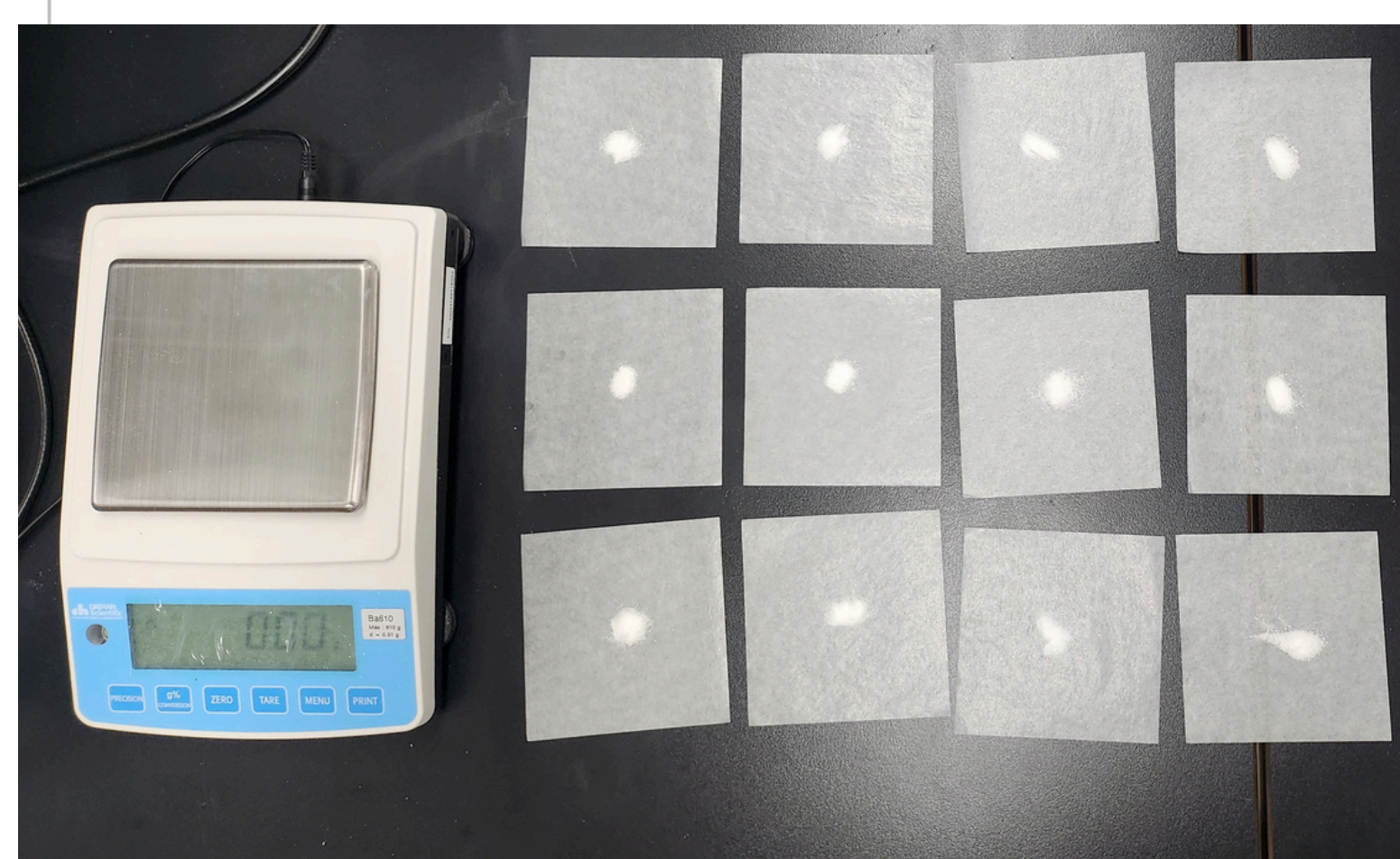
Objectives

To observe meaningful potential for non-invasive blood glucose measurement technology using ultrasound velocity under body fluid conditions like pH 7 and 37°C, the following are needed:

- 1) Confirmation of a linear increase in ultrasound velocity according to glucose concentration.
- 2) The degree of dispersion of ultrasound velocity per glucose concentration.



Experimental Set-up



- Thermometer
- pH7 Standard Buffer Solution
- Quartz Cell, 10.5mL
- LABTron EVO HS1 (Hotplate)
- Stand

$$\text{Concentration} = \frac{m_{\text{solute}}}{V_{\text{solution}}} \times 100 = \frac{m_{\text{solute}}}{800 + \frac{m_{\text{solute}}}{1540}} \times 100 \text{ mg/dL}$$

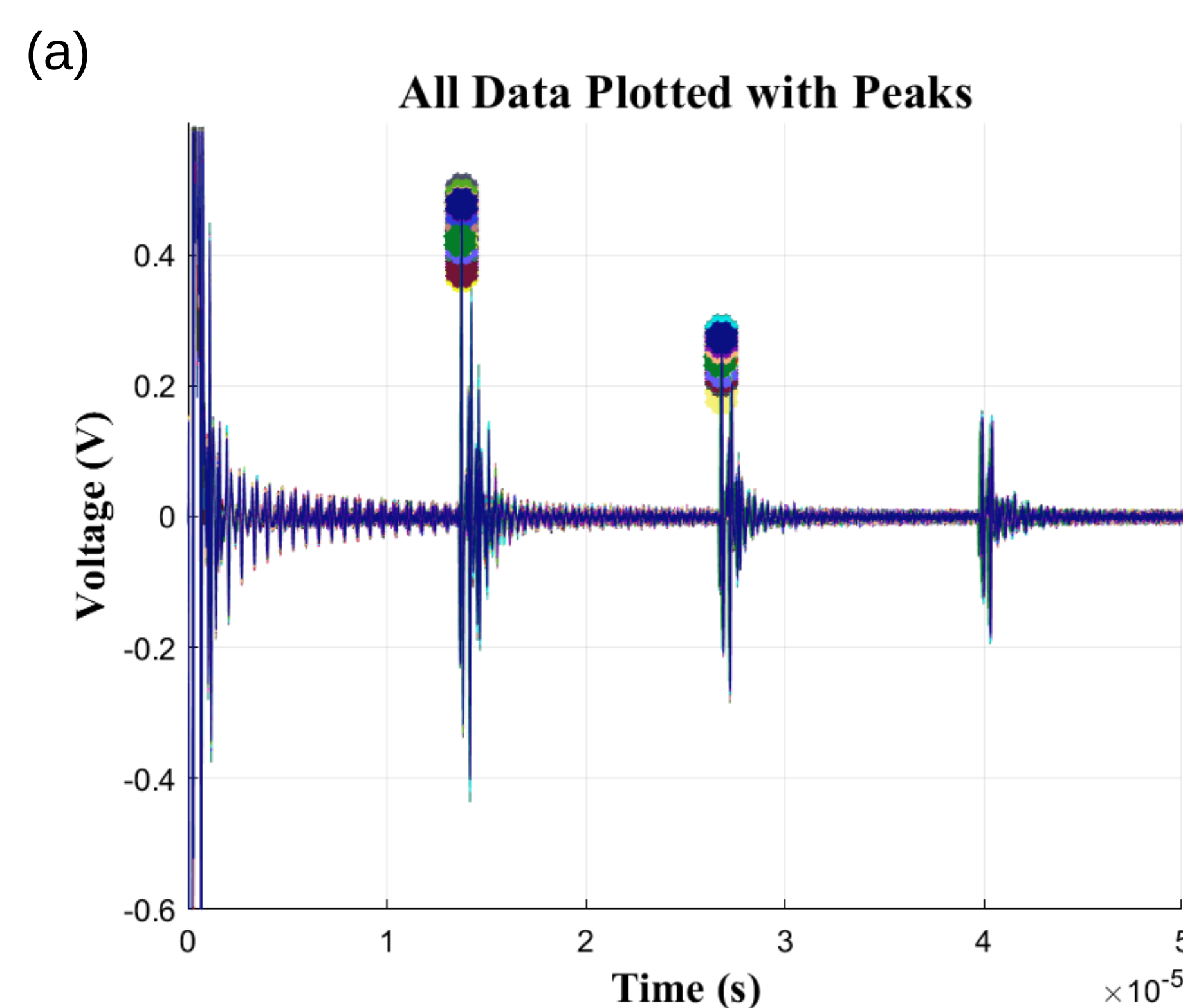
*The density of hydrated glucose, approximately 1.54 g/mL(or 1540 mg/mL)

Glucose (mg)	300	400	500	600	700	800	900	1000	1100	1200
Concentration (mg/dL)	37.49	49.98	62.47	74.96	87.45	99.94	112.42	124.9	137.38	149.85

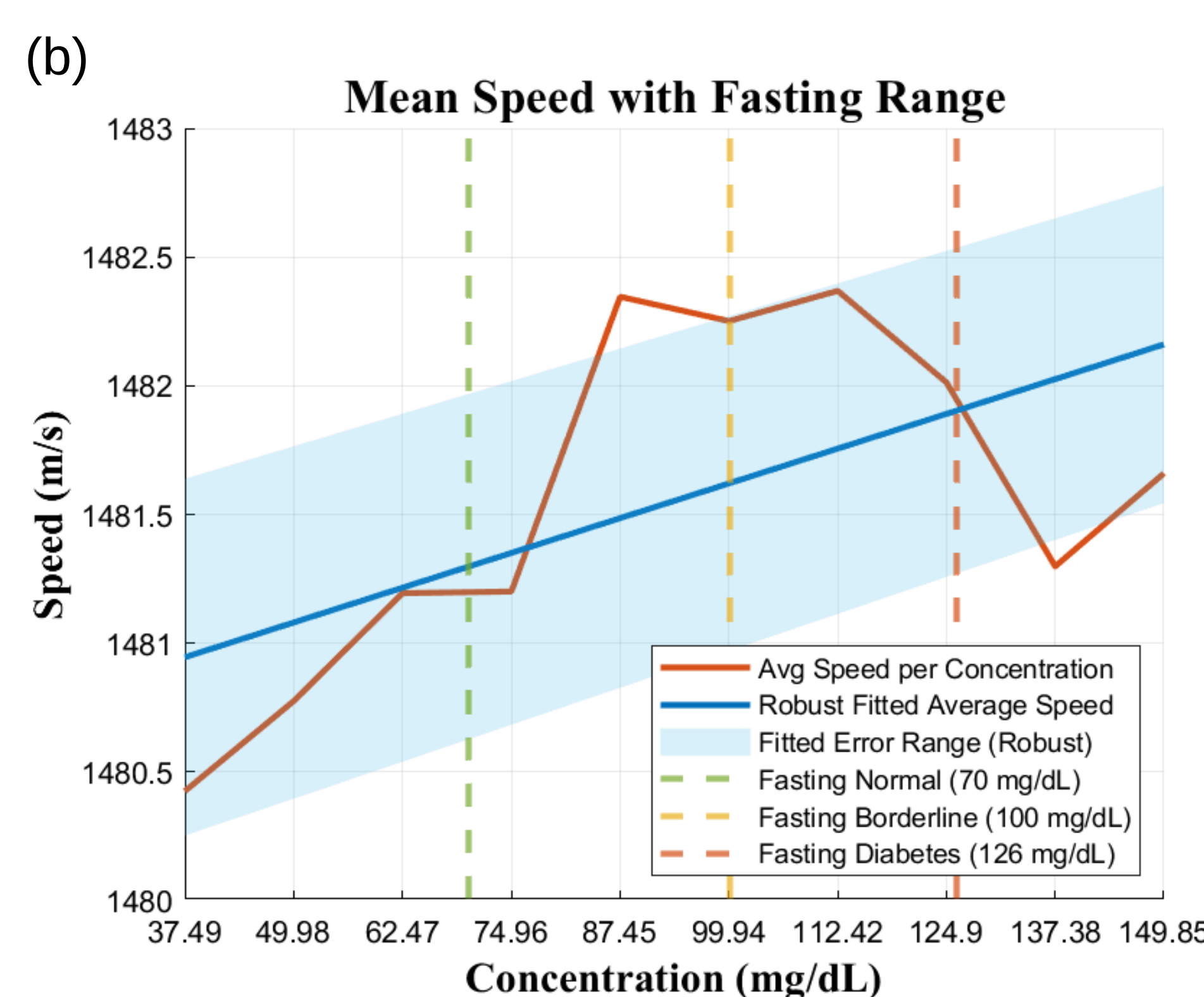
- The volume of the solvent, pH 7 standard buffer solution, is 800mL.
- Twelve 100 mg aliquots were prepared. The experiment started by dissolving 300 mg of solute, with 100 mg added at each step.
- Ten repeated measurements were taken for each concentration.
- The path length of the ultrasonic wave in the quartz cell is 0.01m, so the ultrasonic velocity is calculated based on a round trip distance of 0.02m.
- The transducer operates at a frequency of 5MHz.



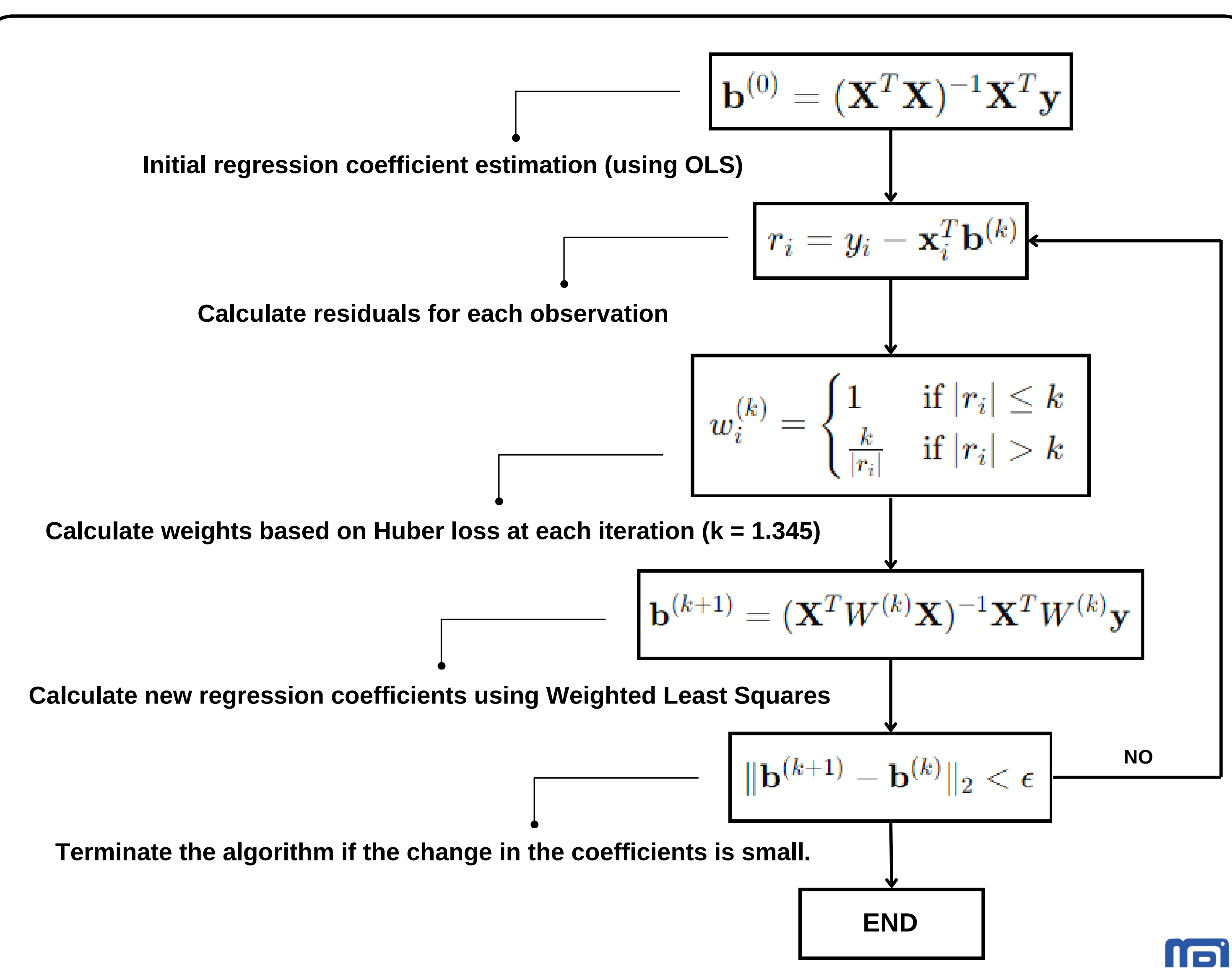
Experimental Results



(a) Speed was calculated using the time difference between the peaks of the first and second packets.



(b) Robust regression was applied using the average speed and standard deviation. This graph shows a linear trend, where the baseline of the '-' marking represents the fasting glucose levels for normal, high-risk, and diabetic groups.



Conclusion

- Based on acidity and temperature, the ultrasonic velocity of glucose solution in a pH 7 standard buffer solution at 37°C shows a linearly increasing trend with glucose concentration.
- **This confirms the potential of ultrasound technology for non-invasive blood glucose measurement. However, the change in ultrasonic velocity in response to meaningful changes in blood glucose concentration was observed to be minimal.**
- Stable and reliable measurements require a measurement system with high accuracy.

